



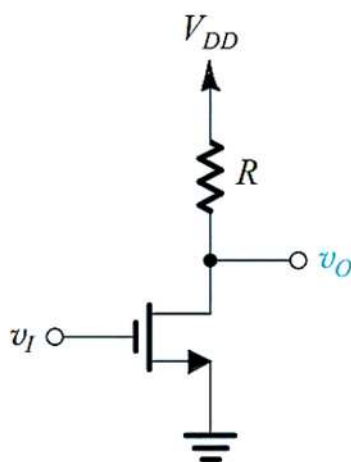
United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Final-Term Exam: Trimester: Spring 2025

Course Code: EEE 2123; Course Title: Electronics
Total Marks: 40; Duration: 2hr

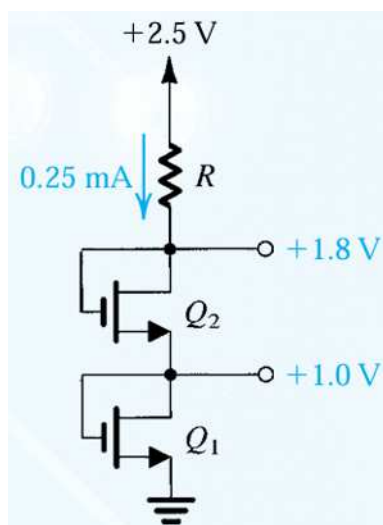
Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

There are six (6) questions in this paper. Answer all of them.

1. The MOSFET in the following figure has $V_t = 0.5\text{V}$, $k'_n = 400 \mu\text{A/V}^2$. Find the required values of W/L and R when $v_I = V_{DD} = +1.8\text{V}$, $r_{DS} = 50\Omega$, and $V_O = 50 \text{ mV}$. [6]

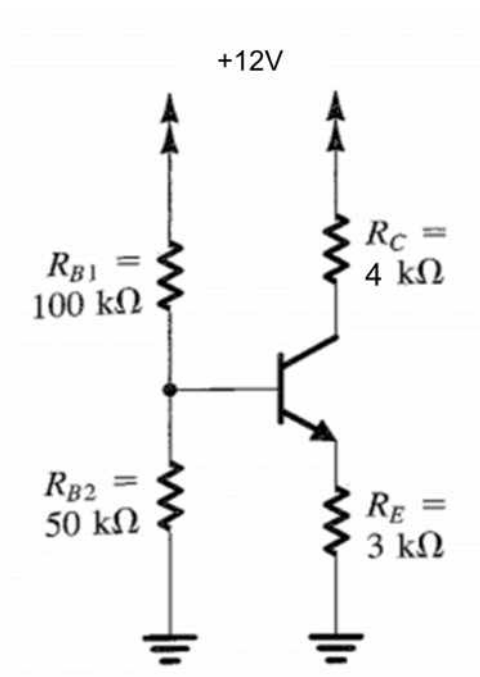


2. The NMOS transistors in the circuit of Fig. P5.50 have $V_t = 0.5 \text{ V}$, $\mu_n C_{ox} = 250 \mu\text{A/V}^2$, and $L_1 = L_2 = 0.25 \mu\text{m}$. Find the required values of gate width for each of Q_1 and Q_2 , and the value of R , to obtain the voltage and current values indicated. [7]



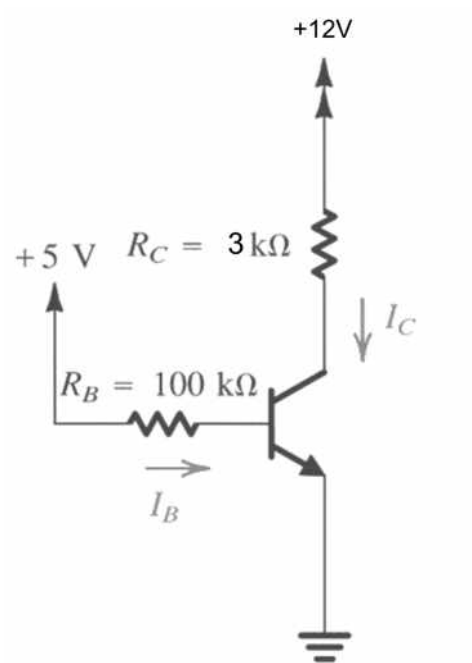
3. For the transistor in the following circuit, $\beta = 50$ and $\beta_{\text{forced}} = 15$. Analyze the circuit to determine the voltage values at all nodes and the current through all branches.

[8]

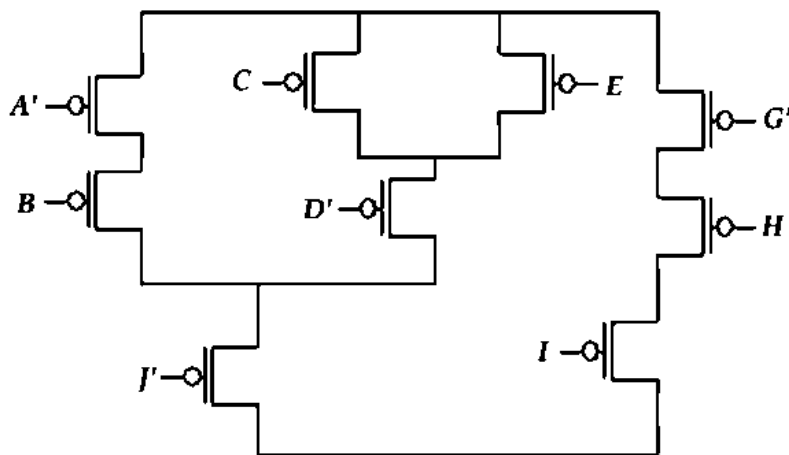


4. Find all node voltages and branch currents in the following circuit, $\beta = 155$

[7]



5. (a) The PMOS network of a random digital system is given below: [4]



Draw the NMOS network and derive the corresponding output expression.

- (b) Implement the following logic function using CMOS: [3]
 $f = (ab + c)ed + gh$

Here are two similar questions that involve implementing mathematical equations using Op-Amps, focusing on integration, differentiation, and summing amplifiers:

- Q6 (a)** Implement the following equation using Op-Amps. Clearly mention the values of all resistances and capacitances used in the design. [4]

$$V_o = 4 \int V_i dt - 15 \frac{dV_i}{dt} + 5V_i$$

In this question, v_i is the input signal. Use appropriate combinations of integrators, differentiators, and gain blocks to implement this.

- Q6 (b)** Design an Op-Amp circuit that implements the following output function. Only $\sin(t)$ is provided as the input signal. [3]

$$V_o = \cos(t) - 3 \sin(t)$$

[Hint: can be obtained by integration. Use suitable scaling with Op-Amps to achieve the coefficients.**]